

# ELECTRIC CIRCUITS · TEACHER GUIDE

Twelve 50-minute lessons, written for students with **no prior knowledge of electricity**. Each lesson has a matching section in the student workbook and a mission on the companion website.

## HOW THE THREE RESOURCES FIT TOGETHER

| RESOURCE                       | USED FOR                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Student workbook</b>        | One booklet per lesson (12 pt text, wide writing lines), each with its toolkit, key words and quick answers, plus a Unit review booklet with the static electricity page (Physics only) and progress tracker. Every lesson: explanations, worked examples, graded practice, lab records, an exam question and a revision box. It is the students' classwork book and their revision guide. |
| <b>Answer edition</b>          | Same pages and page numbers as each booklet, with answers in blue. For marking, and for release to students for self-marking if you choose.                                                                                                                                                                                                                                                |
| <b>Companion website</b>       | panphy.app/gcsephy/year10phy/unit01/: homework and revision. Missions carry more questions with hints and worked answers; the Exam Zone is for the end of the unit.                                                                                                                                                                                                                        |
| <b>Virtual labs, RP sheets</b> | Lessons 2–4 use the Virtual Labs worksheet (PhET) for the full investigation; the workbook holds a short record. Lab 3 plots current against p.d., as AQA does. Lessons 5, 7 and 9 have full required-practical pages on the website.                                                                                                                                                      |

## DESIGN PRINCIPLES

- **Concrete before abstract.** Lesson 1 builds real circuits and teaches symbols and drawing rules before any quantity is measured. The voltmeter appears only when students know what p.d. is (Lesson 3).
- **One new idea at a time.** Current, p.d. and resistance get a lesson each, each with a model (bike chain, delivery vans, collisions) and its limitation. Common misconceptions appear as student claims (Mia, Leo, Ava) to correct.
- **Worked example → your turn → faded practice.** Calculation questions go from a full frame (Equation, Substitute, Answer) to no frame to a challenge.
- **Retrieval every lesson.** 'Do now' questions revisit earlier lessons; 'Revise it' gives cover-and-answer questions with quick answers at the back.
- **Equations are learned.** This cohort sits GCSE in 2028; AQA has confirmed equation sheets only up to 2027, so the booklet teaches recall.

## SEQUENCE

| #  | LESSON                                                                                                                                                       | COMPANION WEBSITE                        |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 1  | <b>Complete circuits and symbols</b><br>What a circuit needs, circuit symbols and drawing rules, measuring current                                           | Mission 1 · Build & measure              |
| 2  | <b>Current and charge</b><br>Current as a rate of flow of charge, $Q = I t$ , current is not used up                                                         | Mission 2 · Charge on the move           |
| 3  | <b>Potential difference</b><br>p.d. as energy per coulomb, voltmeters, $E = Q V$ , p.d.s add in series                                                       | Mission 3 · Energy per coulomb           |
| 4  | <b>Resistance and <math>V = I R</math></b><br>Resistance, $V = I R$ , ohmic conductors, why heating raises resistance                                        | Mission 4 · Resistance & $V = I R$       |
| 5  | <b>Resistance of a wire</b> <b>REQUIRED PRACTICAL</b><br>Required practical: how the length of a wire affects its resistance                                 | Required practical · Resistance          |
| 6  | <b>Series and parallel circuits</b><br>Current, p.d. and resistance rules for series and parallel circuits                                                   | Mission 5 · Series & parallel            |
| 7  | <b>Resistors in series and parallel</b> <b>REQUIRED PRACTICAL</b><br>Required practical: measure total resistance of two resistors in series and in parallel | Required practical · Resistance (Part B) |
| 8  | <b>I–V graphs</b><br>I–V graphs of a fixed resistor, filament lamp and diode                                                                                 | Mission 6 · I–V characteristics          |
| 9  | <b>Investigating I–V characteristics</b> <b>REQUIRED PRACTICAL</b><br>Required practical: measure the I–V characteristics of a resistor, lamp and diode      | Required practical · I–V characteristics |
| 10 | <b>Thermistors, LDRs and sensors</b><br>How thermistors and LDRs respond to temperature and light, and sensor circuits                                       | Mission 6 · Thermistor and LDR           |
| 11 | <b>Power and energy transfer</b><br>Power, $E = P t$ , $P = V I$ , $P = I^2 R$ , energy transfers in appliances                                              | Mission 7 · Power & the National Grid    |
| 12 | <b>Mains, safety and the National Grid</b><br>ac and dc, 230 V 50 Hz mains, three-core cable, earthing, the National Grid                                    | Missions 7 and 8                         |

# COMPLETE CIRCUITS AND SYMBOLS

AQA Combined 6.2.1.1 · Physics 4.2.1.1 · 50 minutes

**BIG QUESTION** Your torch won't switch on. The bulb is fine and the batteries are new. What could be wrong?

## STUDENTS WILL BE ABLE TO

- explain why a lamp only lights in a complete loop
- recognise and draw standard circuit symbols
- turn a real circuit into a neat circuit diagram
- connect an ammeter in series to measure current

## WHY THIS LESSON, HERE

Students start with no formal knowledge. Before any quantity or equation, they need a working idea of a circuit and the vocabulary to draw one. The old Lesson 1 asked for six symbols that were never taught; this lesson teaches them explicitly, from real apparatus to symbol to diagram.

## PREPARATION AND EQUIPMENT

Per pair: 1.5 V cell (or 2 × AA in a holder), lamp in holder, switch, 4 leads, ammeter (0–1 A). One broken torch for the hook (loose battery contact). Optional: a bike or a chain loop to show the model.

## LESSON SEQUENCE

| MIN   | PHASE                             | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON)                                                                                              |
|-------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0–6   | <b>Hook + starting point</b>      | Show the torch that won't work. Students answer the three 'What do you already think?' boxes. Collect ideas; do not correct yet.                     |
| 6–16  | <b>Build first</b>                | Pairs build cell + switch + lamp from the picture, before any symbols. Ask: what did every working circuit have in common? Draw out 'complete loop'. |
| 16–26 | <b>Picture → symbol → diagram</b> | Teach the solid row of symbols and the drawing rules. Model drawing the pairs' own circuit on the board with a ruler.                                |
| 26–32 | <b>Ammeter</b>                    | Add the ammeter in series; read it. Name 'current' and 'series' only. Charge, $Q = I t$ and p.d. wait until Lessons 2–3.                             |
| 32–44 | <b>Try it</b>                     | Questions 1–5. Circulate on Q4 (spot the mistakes) and Q5 (draw it): these show whether the drawing rules have landed.                               |
| 44–50 | <b>Exam corner</b>                | Q7 as a hinge: check symbols, loop and ammeter placement. Q6 (torch detective) for homework if needed.                                               |



**Hinge check.** Show circuit C from Q3. 'Will the lamp light?' If more than a few say yes, reteach 'a gap anywhere stops the current everywhere' before moving on.

## MISCONCEPTIONS TO EXPECT

| STUDENTS MAY THINK...                                                                   | ADDRESS IT BY...                                                                     |
|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| The cell 'fills up' the wire with electricity, which travels from the cell to the lamp. | Use the bike-chain model: the charges are already in the wire and all move together. |
| Current is used up by the lamp.                                                         | Plant the idea with two ammeters now; test it properly in Lesson 2.                  |
| Symbols can go anywhere near the wire.                                                  | Q4 makes the rule explicit: the wire must go into the symbol.                        |

## SUPPORT

Give out a symbol card set to match with real components before drawing. Allow tracing the symbol bank for Q2.

## STRETCH

Draw a circuit with two lamps and a switch that controls only one of them (early preview of parallel).

**Homework:** Companion website, Mission 1: notes and all five questions. Workbook: Revise it.

## CURRENT AND CHARGE

AQA Combined 6.2.1.2 · Physics 4.2.1.2 · Virtual Lab 1 · 50 minutes

**BIG QUESTION** Flick a switch and the lamp lights instantly, yet each electron drifts through the wire slower than a snail. How can both be true?

**STUDENTS WILL BE ABLE TO**

- describe current as the rate of flow of charge
- use  $Q = I t$ , with time in seconds
- explain why the current is the same everywhere in a single loop

**WHY THIS LESSON, HERE**

Current is defined as a rate before students calculate with it. The model (bike chain, then ticket barrier) explains the instant switch-on puzzle and why current is conserved, so  $Q = I t$  is attached to meaning rather than a triangle.

**PREPARATION AND EQUIPMENT**

Computers or tablets with PhET Circuit Construction Kit: DC (Lab). Virtual Labs worksheet pp. 3–6 for pairs who finish. Optional: a real series loop with two ammeters.

**LESSON SEQUENCE**

| MIN   | PHASE                | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON)                                                                              |
|-------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 0–6   | <b>Do now</b>        | Retrieval from Lesson 1. Then pose the big question and take predictions.                                                            |
| 6–18  | <b>Learn</b>         | Free electrons, coulombs, current as a rate (ticket barrier). Model the worked example; students do 'Your turn' on mini whiteboards. |
| 18–32 | <b>Virtual Lab 1</b> | Pairs measure current at three points, then change R. Record in the Lab box. Ask: 'Where did the current go?'                        |
| 32–44 | <b>Try it</b>        | Q1–Q5 in order: the calculation questions fade from full frame to no frame. Q6 is the challenge. Q7 is the misconception check.      |
| 44–50 | <b>Exam corner</b>   | Questions 10–11. Peer mark using the tip: time conversion, unit.                                                                     |



**Hinge check.** Q2 (0.60 A means...). Anyone choosing '0.60 C in the circuit' needs the ticket-barrier model again.

**MISCONCEPTIONS TO EXPECT**

| STUDENTS MAY THINK...                | ADDRESS IT BY...                                                           |
|--------------------------------------|----------------------------------------------------------------------------|
| Current is used up.                  | Virtual Lab 1 plus Q7. Language: charge goes round; energy is transferred. |
| Minutes go straight into $Q = I t$ . | Every supported frame has a 'Convert' row first.                           |
| Electrons flow from + to –.          | Show both arrows on one diagram; exam diagrams use conventional current.   |

**SUPPORT**

Keep the frames; give a units card (C, A, s).

**STRETCH**

How many electrons pass in 1 s at 1 A? ( $\approx 6 \times 10^{18}$ .) Explain why a bigger current does not need faster-moving electrons in a thicker wire.

**Homework:** Mission 2 practice questions (with worked answers). Finish Virtual Lab 1 pp. 5–6 if not done.

## POTENTIAL DIFFERENCE

AQA Combined 6.2.1.3, 6.2.4.2 · Physics 4.2.1.3, 4.2.4.2 · Virtual Lab 2 · 50 minutes

**BIG QUESTION** An AA battery says 1.5 V. A car battery says 12 V. What is that number actually telling you?

## STUDENTS WILL BE ABLE TO

- explain p.d. as the energy transferred per coulomb
- connect a voltmeter in parallel across a component
- use  $E = QV$
- explain why the p.d.s in a loop add up to the supply p.d.

## WHY THIS LESSON, HERE

p.d. is the hardest idea in the unit. It is taught as energy per coulomb, with a model (delivery vans) and an explicit contrast with current, before it is used in  $V = IR$ . The voltmeter is introduced here, when students know what it measures.

## PREPARATION AND EQUIPMENT

PhET (Lab) and Virtual Labs worksheet pp. 7–11. Real demo: 6 V supply, two lamps in series, two voltmeters.

## LESSON SEQUENCE

| MIN   | PHASE         | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON)                                                              |
|-------|---------------|----------------------------------------------------------------------------------------------------------------------|
| 0–6   | Do now        | Retrieval: current, $Q = It$ , current not used up.                                                                  |
| 6–18  | Learn         | Delivery-van model; $1\text{ V} = 1\text{ J/C}$ . Worked example and Your turn. Voltmeter in parallel: say 'across'. |
| 18–30 | Virtual Lab 2 | Measure resistor p.d. at 3, 6, 9 V; calculate $E$ for 2 C. Add a second resistor and test 'p.d.s add up'.            |
| 30–43 | Try it        | Q1 (meter placement) as a class hinge first, then Q2–Q6.                                                             |
| 43–50 | Exam corner   | Questions 9–10; model a two-idea explanation (joules + per coulomb).                                                 |



**Hinge check.** Q1: which circuit measures the p.d. across the lamp? Students who pick B are still thinking of meters as 'in the loop'.

## MISCONCEPTIONS TO EXPECT

| STUDENTS MAY THINK...                              | ADDRESS IT BY...                                                  |
|----------------------------------------------------|-------------------------------------------------------------------|
| Voltage is the speed or 'strength' of the current. | Q6 (Leo). Energy per coulomb, not speed.                          |
| p.d. 'at' a point.                                 | A p.d. is always between two points; insist on 'across'.          |
| Voltage is used up so current falls.               | Energy is transferred; current stays the same (link to Lesson 2). |

## SUPPORT

Current vs p.d. comparison table on a card; sentence starter '\_\_\_ J of energy is transferred for every \_\_\_'.

## STRETCH

A 9 V battery drives 0.50 A for 40 s. Find the charge and then the energy (two-step).

**Homework:** Mission 3 practice questions. Revise it box.

RESISTANCE AND  $V = IR$ 

AQA Combined 6.2.1.3–6.2.1.4 · Physics 4.2.1.3–4.2.1.4 · Virtual Lab 3 · 50 minutes

**BIG QUESTION** A dimmer switch makes a lamp glow less brightly. What is it actually changing?**STUDENTS WILL BE ABLE TO**

- describe resistance and its unit, the ohm ( $\Omega$ )
- use  $V = IR$ , including mA and k $\Omega$
- recognise an ohmic conductor from data or a graph
- explain why heating a metal increases its resistance

**WHY THIS LESSON, HERE**

$V = IR$  now joins current and p.d., which students understand separately. The particle picture of resistance (collisions with ions) is taught here so that the lamp curve in Lesson 8 is a consequence, not a new fact.

**PREPARATION AND EQUIPMENT**

PhET (Lab), Virtual Labs pp. 12–16. Demo: dimmer (variable resistor) with a lamp.

**LESSON SEQUENCE**

| MIN   | PHASE                | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON)                                                     |
|-------|----------------------|-------------------------------------------------------------------------------------------------------------|
| 0–6   | <b>Do now</b>        | Retrieval: p.d. meaning, voltmeter, p.d.s add up.                                                           |
| 6–18  | <b>Learn</b>         | Resistance as opposition; two circuits at 6 V. $V = IR$ with both worked examples (the second includes mA). |
| 18–30 | <b>Virtual Lab 3</b> | Change battery only; $V \div I$ stays at 10 $\Omega$ . Note the axis convention box.                        |
| 30–44 | <b>Try it</b>        | Q1–Q4 (fading calculation support), Q5 table + graph, Q6 temperature.                                       |
| 44–50 | <b>Exam corner</b>   | Questions 9–10.                                                                                             |

**Hinge check.** Q1: same battery, 10  $\Omega$  vs 30  $\Omega$ . Tests the direction of the relationship before any number work.**MISCONCEPTIONS TO EXPECT**

| STUDENTS MAY THINK...                                                   | ADDRESS IT BY...                                     |
|-------------------------------------------------------------------------|------------------------------------------------------|
| Bigger resistance → bigger current ('more resistor, more electricity'). | Two circuits side by side at the same p.d.           |
| Any straight line means ohmic.                                          | Straight line through the origin; check $V \div I$ . |
| mA used as A.                                                           | Worked example 2 converts explicitly.                |

**SUPPORT**

Rearranging help: the companion website's equation triangles (optional). Keep frames.

**STRETCH**

Mission 4 question 5 (electrons and ions). Why does a lamp's resistance rise but a resistor's not (in normal use)?

**Homework:** Mission 4 practice questions.

## RESISTANCE OF A WIRE

AQA Combined RP 15 · Physics RP 3 · Part A · 50 minutes

**BIG QUESTION** If you double the length of a wire, what happens to its resistance?**STUDENTS WILL BE ABLE TO**

- set up a circuit to find resistance from  $V$  and  $I$
- identify and control the variables
- record, process and graph results
- write a conclusion using evidence

**WHY THIS LESSON, HERE**

The first required practical. Students can now draw the circuit, place both meters and calculate  $R$ , so the practical is about method, variables and evidence rather than new content.

**PREPARATION AND EQUIPMENT**

Per group: low-voltage dc supply ( $\approx 2\text{ V}$ ), switch, ammeter, voltmeter, 1.1 m of constantan/nichrome wire (e.g. 0.3 mm) taped to a metre ruler, two crocodile clips, leads. Required practical worksheet (Resistance, Part A) as extension/reference.

**LESSON SEQUENCE**

| MIN   | PHASE            | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON)                                                     |
|-------|------------------|-------------------------------------------------------------------------------------------------------------|
| 0–6   | <b>Do now</b>    | Retrieval: $R = V \div I$ , heating, meter symbols.                                                         |
| 6–14  | <b>Plan</b>      | Complete the variables table (Q1). Walk through the circuit and safety. Stress switch-off between readings. |
| 14–34 | <b>Practical</b> | Ten lengths, 10–100 cm. Readings in the Results table. Repeat any anomaly immediately.                      |
| 34–44 | <b>Process</b>   | Calculate $R$ , plot the graph, line of best fit. Graphs can be finished for homework.                      |
| 44–50 | <b>Analyse</b>   | Q4 and Q5 (pattern and prediction). Exam corner Q9 can be set as homework.                                  |



**Hinge check.** Why switch off between readings? (Q7.) If students answer 'to save the battery', reteach heating  $\rightarrow$  resistance.

**MISCONCEPTIONS TO EXPECT**

| STUDENTS MAY THINK...                                      | ADDRESS IT BY...                                                              |
|------------------------------------------------------------|-------------------------------------------------------------------------------|
| 'Measure resistance with the voltmeter.'                   | Resistance is calculated from $V$ and $I$ .                                   |
| Line of best fit = dot-to-dot.                             | Model a single straight line through the origin region, ignoring the anomaly. |
| Increasing length reduces current, so 'current is the DV'. | The DV is $R$ (calculated).                                                   |

**SUPPORT**

Pre-scaled axes are provided; give a partially completed example table.

**STRETCH**

Q8: explain a non-zero intercept (zero error at the clip).

**Homework:** Finish graph. Exam corner six-mark method. Mission 4 questions 6–9 (measuring resistance and the wire practical).

# SERIES AND PARALLEL CIRCUITS

AQA Combined 6.2.2 · Physics 4.2.2 · 50 minutes

**BIG QUESTION** When one bulb in your kitchen blows, why don't all the other lights in the house go out?

## STUDENTS WILL BE ABLE TO

- tell series and parallel circuits apart
- use the rules for current and p.d. in each
- calculate total resistance in series
- explain why adding a parallel branch lowers the total resistance

## WHY THIS LESSON, HERE

The series/parallel rules are taught as consequences of the Lesson 2–3 ideas: current is conserved (so it splits and rejoins) and energy per coulomb is shared (so p.d.s add in series).

## PREPARATION AND EQUIPMENT

Per pair: cells, three identical lamps, two resistors, ammeters and voltmeters; or PhET. Mini whiteboards.

## LESSON SEQUENCE

| MIN   | PHASE              | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON)                                                       |
|-------|--------------------|---------------------------------------------------------------------------------------------------------------|
| 0–6   | <b>Do now</b>      | Retrieval: $V = IR$ , wire length, meaning of volt.                                                           |
| 6–18  | <b>Learn</b>       | Build two lamps in series, then in parallel; unscrew one lamp in each. Fill the rules table from observation. |
| 18–28 | <b>Measure</b>     | Measure the numbers shown in the two labelled circuits. Check 'adds up' each time.                            |
| 28–44 | <b>Try it</b>      | Worked example + Your turn, then Q1–Q6.                                                                       |
| 44–50 | <b>Exam corner</b> | Question 7.                                                                                                   |



**Hinge check.** Q5: 6  $\Omega$  and 12  $\Omega$  in parallel. Anyone choosing 18  $\Omega$  is adding in parallel.

## MISCONCEPTIONS TO EXPECT

| STUDENTS MAY THINK...                                | ADDRESS IT BY...                              |
|------------------------------------------------------|-----------------------------------------------|
| Current splits equally at every junction.            | Use unequal branch currents (0.25 A, 0.15 A). |
| p.d. splits between parallel branches.               | Each branch has the full supply p.d.          |
| Adding a resistor always increases total resistance. | Checkout model: another route lowers it.      |

## SUPPORT

Rules table on a card beside every question.

## STRETCH

Mission 5 go-further box:  $1/R = 1/R_1 + 1/R_2$  (beyond AQA, clearly labelled).

**Homework:** Mission 5 questions 1–5.

## RESISTORS IN SERIES AND PARALLEL

AQA Combined RP 15 · Physics RP 3 · Part B · 50 minutes

**BIG QUESTION** Two identical resistors, three ways to connect them. Which arrangement lets the most current flow?

**STUDENTS WILL BE ABLE TO**

- measure the total resistance of resistors in series and in parallel
- compare results with  $R_{\text{total}} = R_1 + R_2$
- explain the results using paths and current
- evaluate the measurements

**WHY THIS LESSON, HERE**

The second part of the resistance practical tests the Lesson 6 rules with measurements, and gives evaluation practice (specific error and improvement).

**PREPARATION AND EQUIPMENT**

Per group: low-voltage supply, switch, ammeter, voltmeter, two identical resistors (e.g.  $10\ \Omega$ ), leads. Resistance worksheet Part B for reference.

**LESSON SEQUENCE**

| MIN   | PHASE              | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON)           |
|-------|--------------------|-------------------------------------------------------------------|
| 0–6   | <b>Do now</b>      | Retrieval: series total, parallel rules.                          |
| 6–12  | <b>Predict</b>     | Q1: rank the three arrangements. Record predictions on the board. |
| 12–32 | <b>Practical</b>   | Four arrangements; V across the combination, I from the supply.   |
| 32–44 | <b>Analyse</b>     | Q3–Q6. Compare with predictions.                                  |
| 44–50 | <b>Exam corner</b> | Question 7.                                                       |



**Hinge check.** Before building: where does the voltmeter go for the parallel pair? (Across both.)

**MISCONCEPTIONS TO EXPECT**

| STUDENTS MAY THINK...                           | ADDRESS IT BY...                                                                         |
|-------------------------------------------------|------------------------------------------------------------------------------------------|
| Measure V across one resistor only in parallel. | Across the combination.                                                                  |
| Evaluation: 'human error'.                      | Name a specific error (lead resistance, resolution, heating) and a matching improvement. |

**SUPPORT**

Circuit diagrams printed at each station.

**STRETCH**

Predict the result for three identical resistors in parallel and test it.

**Homework:** Workbook Revise it; Mission 5 question 6.



**BIG QUESTION** Every component has a 'fingerprint': the shape of its I-V graph. Could you identify a component from its graph alone?

#### STUDENTS WILL BE ABLE TO

- read an I-V graph, including negative values
- sketch and recognise the graphs for a fixed resistor, a filament lamp and a diode
- calculate the resistance at a point with  $R = V \div I$
- explain the shape of each graph

#### WHY THIS LESSON, HERE

I-V graphs are taught as 'fingerprints' after students already know  $R = V \div I$ , ohmic behaviour and why heating raises resistance. The graph shapes follow from ideas they already have.

### PREPARATION AND EQUIPMENT

Visualiser. Optional demo: lamp + variable supply with meters, to show resistance rising. Mini whiteboards for sketching.

### LESSON SEQUENCE

| MIN   | PHASE              | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON)                                            |
|-------|--------------------|----------------------------------------------------------------------------------------------------|
| 0–6   | <b>Do now</b>      | Retrieval: ohmic, heating, R at a point.                                                           |
| 6–20  | <b>Learn</b>       | Reading an I-V graph (both quadrants). Build the three shapes one at a time; students sketch each. |
| 20–26 | <b>Explain</b>     | Lamp chain: current → heating → vibrating ions → collisions → R up → less steep.                   |
| 26–44 | <b>Try it</b>      | Q1–Q5. Q3 (card-sort order) makes the causal chain explicit.                                       |
| 44–50 | <b>Exam corner</b> | Question 6: sketch plus explanation.                                                               |

 **Hinge check.** Q1: identify the three unlabelled graphs.

### MISCONCEPTIONS TO EXPECT

| STUDENTS MAY THINK...                                   | ADDRESS IT BY...                                                   |
|---------------------------------------------------------|--------------------------------------------------------------------|
| Resistance = gradient of the I-V graph.                 | Use $R = V \div I$ at a point; a curve's gradient does not give R. |
| The lamp graph flattens because the current 'runs out'. | Heating raises resistance.                                         |
| Diode conducts both ways but less one way.              | Very high resistance in reverse: almost zero current.              |

### SUPPORT

Sketch templates with axes pre-drawn.

### STRETCH

Mission 6 diode threshold question ( $\approx 0.6$  V).

**Homework:** Mission 6 questions 1–2 and 4.

# INVESTIGATING I-V CHARACTERISTICS

AQA Combined RP 16 · Physics RP 4 · 50 minutes

**BIG QUESTION** Can you produce the fingerprint graphs of a resistor, a lamp and a diode yourself?

**STUDENTS WILL BE ABLE TO**

- set up a circuit to vary the p.d. and measure the current
- take readings in both directions, safely
- plot an I-V graph with positive and negative values
- describe how the method changes for a diode

**WHY THIS LESSON, HERE**

Students now know what each graph should look like, so the practical tests predictions and focuses on method: varying p.d., reversing, safety and the diode's extra steps.

**PREPARATION AND EQUIPMENT**

Per group: variable low-voltage supply or supply + variable resistor, ammeter (and milliammeter), voltmeter, fixed resistor, 6 V filament lamp, diode + protective resistor (e.g. 100  $\Omega$ ), leads. I-V worksheet for reference.

**LESSON SEQUENCE**

| MIN   | PHASE                  | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON)          |
|-------|------------------------|------------------------------------------------------------------|
| 0–6   | <b>Do now + safety</b> | Retrieval. Lamp rating, hot lamp, protective resistor.           |
| 6–12  | <b>Set up</b>          | Check ammeter in series and voltmeter across the component only. |
| 12–34 | <b>Practical</b>       | Resistor and lamp in both directions; diode with small steps.    |
| 34–44 | <b>Graph</b>           | Both components on one centred graph; sketch the diode.          |
| 44–50 | <b>Analyse</b>         | Q4–Q8; exam corner six-mark method as homework.                  |



**Hinge check.** How do we get the negative readings? (Swap the supply connections; record as negative.)

**MISCONCEPTIONS TO EXPECT**

| STUDENTS MAY THINK...                               | ADDRESS IT BY...       |
|-----------------------------------------------------|------------------------|
| Plotting reversed readings as positive.             | Q8.                    |
| Voltmeter across the diode and protective resistor. | Across the diode only. |

**SUPPORT**

Given target p.d.s in the table.

**STRETCH**

Explain why the lamp's resistance at 1 V is still larger than at 0.1 V.

**Homework:** Exam corner six-mark question. Mission 6 question 4.

## THERMISTORS, LDRS AND SENSORS

AQA Combined 6.2.1.4, 6.2.2 · Physics 4.2.1.4, 4.2.2 · 50 minutes

**BIG QUESTION** How does a streetlight know it's getting dark, and how does a fire alarm know the room is hot?

**STUDENTS WILL BE ABLE TO**

- describe how a thermistor's resistance changes with temperature
- describe how an LDR's resistance changes with light intensity
- predict the current in a sensor circuit
- choose the right sensor for a job

**WHY THIS LESSON, HERE**

Thermistors and LDRs are taught as applications of 'lower total  $R \rightarrow$  bigger current at fixed p.d.', so they revise  $V = IR$  and series circuits while adding two new components.

**PREPARATION AND EQUIPMENT**

Thermistor and LDR with multimeter (resistance range); beaker of warm water; torch; mini whiteboards.

**LESSON SEQUENCE**

| MIN   | PHASE                  | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON)                                       |
|-------|------------------------|-----------------------------------------------------------------------------------------------|
| 0–6   | <b>Do now</b>          | Retrieval: lamp graph, $V = IR$ direction, diode symbol.                                      |
| 6–18  | <b>Learn + demo</b>    | Measure $R$ of the LDR (cover/torch) and thermistor (hand/warm water). Sketch the two graphs. |
| 18–28 | <b>Sensor circuits</b> | LDR in series; worked example and Your turn.                                                  |
| 28–44 | <b>Try it</b>          | Q1–Q5.                                                                                        |
| 44–50 | <b>Exam corner</b>     | Question 6.                                                                                   |



**Hinge check.** Q5: which change increases the current through a thermistor?

**MISCONCEPTIONS TO EXPECT**

| STUDENTS MAY THINK...                       | ADDRESS IT BY...                               |
|---------------------------------------------|------------------------------------------------|
| Brighter light increases LDR resistance.    | Both are 'more $\rightarrow$ less'.            |
| Changing resistance changes the supply p.d. | The supply p.d. is fixed; the current changes. |

**SUPPORT**

The 'more  $\rightarrow$  less' key box as a card.

**STRETCH**

Explain why the p.d. across the fixed resistor rises as the LDR is lit (qualitative potential divider).

**Homework:** Mission 6 questions 3 and 5.

## POWER AND ENERGY TRANSFER

AQA Combined 6.2.4.1–6.2.4.2 · Physics 4.2.4.1–4.2.4.2 · 50 minutes

**BIG QUESTION** Which transfers more energy: a 3 kW kettle boiling for 3 minutes, or a 10 W LED lamp left on for 5 hours?

## STUDENTS WILL BE ABLE TO

- describe power as the energy transferred per second
- use  $E = P t$ ,  $P = V I$  and  $P = I^2 R$  with the right units
- describe the energy transfers in everyday appliances
- choose the right equation for a question

## WHY THIS LESSON, HERE

Power pulls together earlier quantities. Students choose between equations (the decision table) instead of memorising a list, and  $P = I^2 R$  prepares the National Grid argument in Lesson 12.

## PREPARATION AND EQUIPMENT

Appliance rating plates (photos or real). Calculators.

## LESSON SEQUENCE

| MIN   | PHASE                | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON)                         |
|-------|----------------------|---------------------------------------------------------------------------------|
| 0–6   | Do now               | Retrieval: $E = Q V$ , $Q = I t$ , $V = I R$ .                                  |
| 6–18  | Learn                | Power as J per s; appliance table; $E = P t$ worked example (unit conversions). |
| 18–28 | More power equations | $P = V I$ and $P = I^2 R$ worked examples; 'square the current first'.          |
| 28–44 | Try it               | Q1 is the big question; then Q2–Q6.                                             |
| 44–50 | Exam corner          | Question 7 (six marks).                                                         |



**Hinge check.** Which equation? Show three sets of data and ask students to name the equation only.

## MISCONCEPTIONS TO EXPECT

| STUDENTS MAY THINK...                                          | ADDRESS IT BY...             |
|----------------------------------------------------------------|------------------------------|
| $P = I^2 R$ calculated as $(I R)^2$ or $I \times 2 \times R$ . | Square first, then multiply. |
| kW and minutes substituted directly.                           | Convert row in every frame.  |

## SUPPORT

Decision table on a card.

## STRETCH

Q6 (lamp current and resistance); cost of energy if the teacher adds kWh.

**Homework:** Mission 7 questions 1–4.

## MAINS, SAFETY AND THE NATIONAL GRID

AQA Combined 6.2.3, 6.2.4.3 · Physics 4.2.3, 4.2.4.3 · 50 minutes

**BIG QUESTION** Pylon cables carry up to 400 000 V. Why so high, and why do we only get 230 V at the plug?

**STUDENTS WILL BE ABLE TO**

- describe the difference between ac and dc
- state the UK mains values: 230 V, 50 Hz
- explain how the earth wire and fuse protect you
- explain why the National Grid uses very high p.d.s

**WHY THIS LESSON, HERE**

Mains safety and the National Grid close the unit as one story, from power station to plug. The grid explanation uses  $P = VI$  and  $P = I^2 R$  from Lesson 11.

### PREPARATION AND EQUIPMENT

Cut-away three-core cable or plug (no mains work by students). Transformer demo if available.

### LESSON SEQUENCE

| MIN   | PHASE                       | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON)                 |
|-------|-----------------------------|-------------------------------------------------------------------------|
| 0–6   | <b>Do now</b>               | Retrieval: $P = VI$ , $P = I^2 R$ , units.                              |
| 6–16  | <b>ac/dc and mains</b>      | Traces or diagrams; 230 V, 50 Hz; cable colours and jobs.               |
| 16–26 | <b>Safety story</b>         | Live danger; earth + fuse sequence. Students order the events (Q3).     |
| 26–36 | <b>National Grid</b>        | Grid diagram; Q5 calculation shows the $\times 10 / \times 100$ effect. |
| 36–50 | <b>Try it + exam corner</b> | Q1–Q6, then exam corner Questions 7–8. Set the Unit review for home.    |



**Hinge check.** Q4 (Ava): is the live wire dangerous when switched off?

### MISCONCEPTIONS TO EXPECT

| STUDENTS MAY THINK...                            | ADDRESS IT BY...                             |
|--------------------------------------------------|----------------------------------------------|
| Earth wire carries current all the time.         | Only during a fault.                         |
| Step-up transformers make extra energy.          | Power in = power out; p.d. up, current down. |
| Neutral is 'safe' and live only matters when on. | Live is at ~230 V whenever connected.        |

### SUPPORT

Fault story as cut-out cards.

### STRETCH

Higher-tier transformer question (Q6); explain why ac is needed for transformers.

**Homework:** Unit review (workbook), then Exam Zone on the companion website.